

\*Equal contribution

## MOTIVATION



**Vision-based control is hard:** robots relying on camera observations can make unsafe decisions such as collisions in real-world environments.

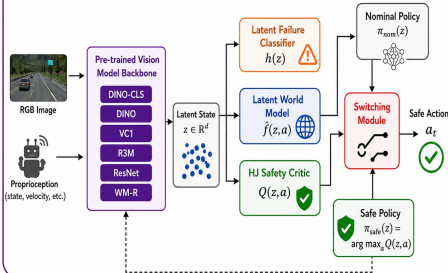


**Safety is critical:** safety filters are mechanisms that monitor reference controller online and minimally intervene to prevent unsafe behavior while preserving task performance.

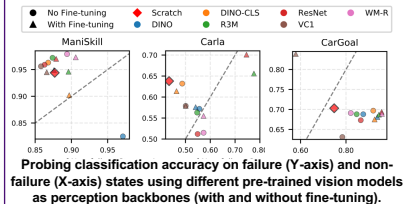


**Pre-trained vision representations (PVRs):** PVRs have shown strong performance for robot control, but their suitability for learning reliable vision-based safety filters remains unclear.

## METHODOLOGY



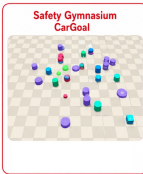
## RESULTS



|                            | DINO-CLS | DINO | VC1  | R3M  | Resnet | WM-R |
|----------------------------|----------|------|------|------|--------|------|
| <b>Inference Time (ms)</b> | 16.1     | 19.3 | 16.7 | 16.3 | 8.9    | 11.9 |
| <b>Model Size (MB)</b>     | 291      | 510  | 611  | 275  | 275    | 275  |

**Inference time and model size for safety filters with different backbones. Model size includes the world model (encoders and predictor) and the safety filter.**

## EXPERIMENTAL SETUP



## TAKEAWAYS



Fine-tuning pre-trained vision models greatly improves safety filtering performance compared to training from scratch.



DINO-CLS consistently achieves the best safety performance across diverse tasks.



Features suited for strong failure classifiers do not necessarily lead to reliable safety filters.



Vision models that excel at general robotic tasks are not necessarily optimal for safety-critical applications.



Latent safety filters are lightweight enough for real-time robotic deployment.

| PVR      | ManiSkill   |             | Dubins      |             | CarGoal      |             |
|----------|-------------|-------------|-------------|-------------|--------------|-------------|
|          | Vio.        | Succ.       | Vio.        | Succ.       | Vio.         | Succ.       |
| DINO-CLS | <b>1.28</b> | <b>0.34</b> | 4.96        | 0.98        | 48.62        | 0.44        |
| DINO     | 3.52        | 0.10        | 9.80        | 0.98        | 57.70        | 0.20        |
| VC1      | 4.76        | 0.02        | 8.92        | 0.98        | 53.48        | 0.38        |
| R3M      | 1.88        | 0.38        | 8.44        | 0.98        | 73.42        | 0.86        |
| ResNet   | 5.40        | 0.32        | 8.76        | 0.98        | 54.82        | 0.42        |
| WM-R     | 3.82        | 0.18        | <b>4.66</b> | <b>0.98</b> | <b>48.26</b> | <b>0.32</b> |
| DINO-CLS | <b>0.42</b> | <b>0.44</b> | <b>0.60</b> | <b>0.98</b> | 30.68        | 0.26        |
| DINO     | 4.62        | 0.36        | 2.58        | 0.98        | 35.78        | 0.26        |
| VC1      | 1.74        | 0.00        | 7.26        | 0.98        | <b>19.40</b> | <b>0.24</b> |
| R3M      | 0.44        | 0.50        | 7.90        | 0.98        | 48.52        | 0.32        |
| ResNet   | 5.30        | 0.32        | 5.50        | 0.98        | 19.74        | 0.20        |
| WM-R     | 8.10        | 0.42        | 4.50        | 0.98        | 35.70        | 0.06        |

**Success rates (Succ.) and total violations (Vio.) for safety filters with different backbones across tasks.**